BOUNDED JENSEN MEASURES ON THE INFINITE POLYDISC AND EXTENSION OF NEVANLINNA FUNCTIONS TO ℓ^2

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Abstract. Guo, Ni and Zhou asked whether every function in the Smirnov class $N_*(\mathbb{T}^\infty)$ extends holomorphically to $\mathbb{D}^\infty \cap \ell^2$. We show that it does, and that the same holds for the Nevanlinna class $N(\mathbb{D}_1^\infty)$, with the explicit bound $\log(1+|F(\zeta)|) \le K(\zeta)\|F\|_0$, where $K(\zeta) \le \exp(2\|\zeta\|_2 + 2\|\zeta\|_2^2/(1-\|\zeta\|_\infty))$. The domain $\mathbb{D}^\infty \cap \ell^2$ is maximal. The proof replaces the Poisson kernel, which is bounded only at points of ℓ^1, by a Jensen measure whose density is bounded independently of the dimension. The same construction shows that a point of the infinite polydisc has a positive representing measure with bounded density with respect to Haar measure if and only if it lies in ℓ^2, and that this measure can be chosen to be a Jensen measure. This answers a question of Cole and Gamelin. As a further consequence, point evaluation on the Nevanlinna class of Dirichlet series is bounded in the half-plane $\operatorname{Re} s > 1/2$, and this half-plane is optimal.

1. Introduction

Let $\mathbb{D}$ be the open unit disc and $\mathbb{T}$ the unit circle, and let m be the Haar measure of the infinite torus $\mathbb{T}^\infty$. Following Guo, Ni and Zhou [4], we write $\mathcal{P}_\infty$ for the analytic polynomials in finitely many variables,

$$\mathbb{D}_1^\infty = \mathbb{D}^\infty \cap \ell^1 \qquad \text{and} \qquad \mathbb{D}_2^\infty = \mathbb{D}^\infty \cap \ell^2,$$

regarded as open subsets of the Banach spaces ℓ^1 and ℓ^2. For a measurable function f on $\mathbb{T}^\infty$ put

$$\|f\|_0 = \int_{\mathbb{T}^\infty} \log(1+|f|)\,\mathrm{d}m.$$

The Smirnov class $N_*(\mathbb{T}^\infty)$ is the closure of $\mathcal{P}_\infty$ in the metric $\|f-g\|_0$. The Nevanlinna class $N(\mathbb{D}_1^\infty)$ consists of the holomorphic functions F on $\mathbb{D}_1^\infty$ with

$$\|F\|_0 = \sup_{0<r<1} \int_{\mathbb{T}^\infty} \log(1+|F_{[r]}|)\,\mathrm{d}m < \infty,$$

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where $F_{[r]}(w) = F(rw_1, r^2w_2, r^3w_3, \dots)$; see [4, Section 2.1]. A function on an open subset of a Banach space is holomorphic if it is locally bounded and holomorphic on each complex line.

For $\zeta \in \mathbb{D}^\infty$, let

$$P_\zeta(w) = \prod_{j\geq 1} \frac{1-|\zeta_j|^2}{|1-\bar{\zeta}_j w_j|^2}, \qquad \|P_\zeta\|_\infty = \prod_{j\geq 1} \frac{1+|\zeta_j|}{1-|\zeta_j|},$$

be the Poisson kernel and its supremum, which is finite exactly when $\zeta \in \ell^1$. Guo, Ni and Zhou proved that

$$\log(1+|F(\zeta)|) \leq \|P_\zeta\|_\infty \|F\|_0, \qquad F \in N(\mathbb{D}_1^\infty),\ \zeta \in \mathbb{D}_1^\infty, \tag{1}$$

see [4, Lemma 2.10]. Applied to polynomials, (1) shows that the point evaluations at $\zeta \in \mathbb{D}_1^\infty$ extend continuously to $N_*(\mathbb{T}^\infty)$, and every $f \in N_*(\mathbb{T}^\infty)$ defines a holomorphic function $\tilde{f}$ on $\mathbb{D}_1^\infty$ [4, Proposition 2.25 and Theorem 2.27]. For $0 < p < \infty$, the functions in $H^p(\mathbb{T}^\infty) \subset N_*(\mathbb{T}^\infty)$ extend holomorphically to the larger domain $\mathbb{D}_2^\infty$, by the work of Cole and Gamelin [3, Theorems 8.1 and 10.1]. This led Guo, Ni and Zhou to ask the following [4, Question 2.28].

Question 1 (Guo, Ni and Zhou)**.** *Does there exist a function in $N_*(\mathbb{T}^\infty)$ which can not be extended to a function holomorphic on $\mathbb{D}_2^\infty$?*

The main purpose of the present paper is to show that the answer is no. For $\zeta \in \mathbb{D}_2^\infty$ we set

$$K(\zeta) = \exp\Big(2\|\zeta\|_2 + \sum_{j\geq 1} \frac{2|\zeta_j|^2}{1-|\zeta_j|}\Big) \leq \exp\Big(2\|\zeta\|_2 + \frac{2\|\zeta\|_2^2}{1-\|\zeta\|_\infty}\Big). \tag{2}$$

Theorem 1. *Let K be as in* (2)*.*

1. *Suppose that $q \in \mathcal{P}_\infty$ and $\zeta \in \mathbb{D}_2^\infty$. Then $\log(1+|q(\zeta)|) \leq K(\zeta)\|q\|_0$.*
2. *Suppose that $f \in N_*(\mathbb{T}^\infty)$. Then $\tilde{f}$ has a unique holomorphic extension F to $\mathbb{D}_2^\infty$, and $\log(1+|F(\zeta)|) \leq K(\zeta)\|f\|_0$ for every $\zeta \in \mathbb{D}_2^\infty$.*
3. *Suppose that $F \in N(\mathbb{D}_1^\infty)$. Then F has a unique holomorphic extension to $\mathbb{D}_2^\infty$, and this extension satisfies $\log(1+|F(\zeta)|) \leq K(\zeta)\|F\|_0$ for every $\zeta \in \mathbb{D}_2^\infty$.*

The domain $\mathbb{D}_2^\infty$ cannot be enlarged: at a point $\zeta \in \mathbb{D}^\infty \setminus \ell^2$ the evaluation $q \mapsto q(\zeta)$ is not continuous on $\mathcal{P}_\infty$ for the metric of $N_*(\mathbb{T}^\infty)$ (Proposition 1 below). In particular, the Smirnov class has the same maximal domain of holomorphic extension as the Hardy spaces.

The constant $K(\zeta)$ replaces the factor $\|P_\zeta\|_\infty$ in (1). Since $\log\|P_\zeta\|_\infty \geq 2\|\zeta\|_1$, this is an improvement on $\mathbb{D}_1^\infty$ when $\|\zeta\|_2$ is much smaller than $\|\zeta\|_1$, for instance at points with many small coordinates. It is not an improvement in general: for $\zeta = (s, 0, 0, \dots)$ with $0 < s < 1$ we have

$$\log K(\zeta) = 2s + \frac{2s^2}{1-s} > \log\frac{1+s}{1-s} = \log\|P_\zeta\|_\infty.$$

On $\mathbb{D}_1^\infty$ both bounds hold, so (1) and Theorem 1 hold with the constant $\min\{K(\zeta), \|P_\zeta\|_\infty\}$.

Representing measures with bounded density. The proof of Theorem 1 rests on Jensen measures with bounded density, and these answer a question of Cole and Gamelin [3]. Let A be the infinite polydisc algebra, that is, the uniform closure of $\mathcal{P}_\infty$ on $\overline{\mathbb{D}}^\infty$ with the product topology. A representing measure for a point $\zeta \in \overline{\mathbb{D}}^\infty$ is a probability measure μ on $\overline{\mathbb{D}}^\infty$ such that $f(\zeta) = \int f \,\mathrm{d}\mu$ for every $f \in A$. It is a Jensen measure if $\log|f(\zeta)| \le \int \log|f| \,\mathrm{d}\mu$ for every $f \in A$. Every Jensen measure is a representing measure.

For $\zeta \in \mathbb{D}_2^\infty$ the Poisson measure $P_\zeta \,\mathrm{d}m$ is a Jensen measure, absolutely continuous with a density in $L^p(m)$ for every $p < \infty$ ([3, Theorem 3.1 and Lemma 3.3]), and the density is bounded if and only if $\zeta \in \ell^1$ [3, Theorem 4.1]. Cole and Gamelin proved that, for fixed $1 < p < \infty$, a point $\zeta \in \mathbb{D}^\infty$ has a representing measure with density in $L^p(m)$ if and only if $\zeta \in \ell^2$ ([3, Section 1], with Theorems 3.1 and 8.1 there). They wrote that they were not able to determine which points have representing measures in $L^\infty(m)$ [3, p. 113]. At the end of [3, Section 8], on p. 130, they asked a sharper question: every $\zeta \in \mathbb{D}_2^\infty$ has a complex representing measure $G \,\mathrm{d}m$ with $|G|$ constant, and they asked whether there is a positive representing measure for ζ with bounded density. Our second result answers both questions.

Theorem 2. *Suppose that $\zeta \in \mathbb{D}^\infty$. Then ζ has a representing measure of the form $k \,\mathrm{d}m$ with $k \in L^\infty(m)$ if and only if $\zeta \in \ell^2$. Moreover, if $\zeta \in \mathbb{D}_2^\infty$, then there is a Jensen measure $k \,\mathrm{d}m$ for ζ with $0 \le k \le K(\zeta)$, where K is as in* (2).

Points $\zeta \in \overline{\mathbb{D}}^\infty$ with $|\zeta_j| = 1$ for some j have no representing measure that is absolutely continuous with respect to m (see the end of Section 3), so Theorem 2 describes all points of $\overline{\mathbb{D}}^\infty$ with a representing measure in $L^\infty(m)$. Cole and Gamelin asked the question on p. 130 also for the product measures $\mathrm{d}Q = \mathrm{d}\omega \times \mathrm{d}\omega \times \cdots$, where ω is a rotation-invariant probability measure on $\overline{\mathbb{D}}$. We treat only the Haar measure m, and the general product measures are not addressed here.

Dirichlet series. Theorem 1 has a consequence for Dirichlet series. Let $\mathscr{N}_u$ be the Nevanlinna class of Dirichlet series of Brevig and Perfekt [2], studied further by Guo, Ni and Zhou [4, Section 4] and by Allu and Mondal [1], and let $\mathscr{N} \supset \mathscr{N}_u$ be its completion. Write $\mathbb{C}_\theta = \{s \in \mathbb{C} : \operatorname{Re} s > \theta\}$. Point evaluation on $\mathscr{N}_u$ (hence on $\mathscr{N}$) is known to be bounded in $\mathbb{C}_1$: this follows from (1), as observed in [1, Section 4, (4.1) and (4.2)], and the constant is $\|P_{r(s)}\|_\infty = \zeta(\sigma)^2/\zeta(2\sigma)$, where $\sigma = \operatorname{Re} s$, ζ is the Riemann zeta function, and $r(s)$ is the point $(p_j^{-s})_{j\ge1}$ indexed by the primes. In Corollary 1 below we extend this to $\mathbb{C}_{1/2}$, with the bound

$$\log(1+|f(s)|) \le \exp\Bigl(2\sqrt{\log\zeta(2\sigma)} + \frac{2\log\zeta(2\sigma)}{1-2^{-\sigma}}\Bigr)\|f\|_0, \qquad \sigma = \operatorname{Re} s > \frac{1}{2},$$

and we show that $\mathbb{C}_{1/2}$ is optimal.

Our approach. Let us briefly describe our approach. The proof of (1) uses that the Poisson measure $P_\zeta \,\mathrm{d}m$ is a Jensen measure for ζ, so that $\log(1+|F(\zeta)|) \le \int \log(1+|F|) P_\zeta \,\mathrm{d}m$, and then bounds P_ζ by its supremum. The function $\log(1+|F|)$ is controlled only in $L^1(m)$, so this step needs a Jensen measure with bounded density, and the Poisson measure has one only at points of ℓ^1. We construct, for every $\zeta \in \mathbb{D}^n$, a different Jensen measure $k \,\mathrm{d}m_n$ on $\mathbb{T}^n$ with $k \le K(\zeta)$, independently of n. It is obtained from the Poisson measure by averaging over the circles $\theta \mapsto e^{i\theta}x$ in $\mathbb{T}^n$: each holomorphic disc $\lambda \mapsto \mu_\zeta(\lambda x)$, where μ_ζ is the coordinatewise Möbius map sending 0 to ζ, passes through ζ, and the averaging removes the dependence of the density on $\|\zeta\|_1$. What remains is controlled by $\sum_j \bar\zeta_j x_j$, whose $L^2(m_n)$ norm is $\|\zeta\|_2$.

Jensen measures for the infinite polydisc algebra were studied by Cole and Gamelin in [3, Section 5]. If the coordinates of ζ tend to zero slowly, then they bound the absolutely continuous part of every Jensen measure for ζ [3, Theorem 5.3], and they give points of c_0 at which every Jensen measure is singular to m [3, Theorem 5.4]. A density in L^p for $p < \infty$ suffices for Hardy spaces, by Hölder's inequality, but not for the Nevanlinna and Smirnov classes, since $\int \log(1+|f|)^{p'} \,\mathrm{d}m$ is not controlled by $\|f\|_0$ for any $p' > 1$. To the best of our knowledge, Lemma 1, Theorem 1 and Theorem 2 are new, and Theorem 2 answers the questions of Cole and Gamelin [3, pp. 113 and 130] for the Haar measure m.

Organization. The present paper is organized into six additional sections. In Section 2 we fix notation. In Section 3 we construct the Jensen measures with bounded density and prove Theorem 2. In Section 4 we prove Theorem 1, and Section 5 shows that $\mathbb{D}_2^\infty$ is maximal. In Section 6 we deduce the result on Dirichlet series. In Section 7 we discuss other Jensen measures, the dependence of the constant on ζ, and some open problems.

2. Preliminaries

We write m_n for the Haar measure of $\mathbb{T}^n$, and we identify a function on $\mathbb{T}^n$ with a function on $\mathbb{T}^\infty$ that depends on the first n variables only. For $\zeta \in \mathbb{D}^\infty$ and $n \ge 1$ we let $\zeta^{(n)} = (\zeta_1, \dots, \zeta_n, 0, 0, \dots)$. For $0 < r < 1$ and a sequence ζ we write $r \circ \zeta = (r^j \zeta_j)_{j \ge 1}$, so that $F_{[r]}(w) = F(r \circ w)$.

For $a \in \mathbb{D}$ let $\mu_a(z) = (z+a)/(1+\bar a z)$. Then μ_a is an automorphism of the closed disc $\overline{\mathbb{D}}$ which maps $\mathbb{T}$ onto $\mathbb{T}$, with $\mu_a(0) = a$, and

$$|\mu_a'(x)| = J_a(x) \coloneqq \frac{1-|a|^2}{|1+\bar a x|^2}, \qquad x \in \mathbb{T}.$$

For $\zeta \in \mathbb{D}^n$ and $x \in \overline{\mathbb{D}}^n$ we put $\mu_\zeta(x) = (\mu_{\zeta_1}(x_1), \dots, \mu_{\zeta_n}(x_n))$ and, for $x \in \mathbb{T}^n$, $J(x) = \prod_{j=1}^n J_{\zeta_j}(x_j)$. The change of variables $y = \mu_\zeta(x)$ gives

$$\int_{\mathbb{T}^n} h(\mu_\zeta(x)) J(x) \, \mathrm{d}m_n(x) = \int_{\mathbb{T}^n} h \, \mathrm{d}m_n \tag{3}$$

for every nonnegative Borel function h on $\mathbb{T}^n$.

We will use the following special case of [4, Corollary 2.2]. Suppose that u is continuous on $\overline{\mathbb{D}}^\infty$, with the product topology, and subharmonic in each variable on $\mathbb{D}_1^\infty$. Then

$$u(0) \le \int_{\mathbb{T}^\infty} u \, \mathrm{d}m. \tag{4}$$

3. Jensen measures with bounded density

The next result is the key to Theorems 1 and 2. It provides, at every point of $\mathbb{D}^n$, a Jensen measure on $\mathbb{T}^n$ whose density is bounded in terms of the ℓ^2 norm of the point, independently of n.

Lemma 1. *Suppose that $n \ge 1$ and $\zeta \in \mathbb{D}^n$, and let $\tau = \|\zeta\|_2$. Then there is a Borel function k on $\mathbb{T}^n$ with $\int k \, \mathrm{d}m_n = 1$ and*

$$0 \le k \le \exp\Big(2\tau + \sum_{j=1}^n \frac{2|\zeta_j|^2}{1 - |\zeta_j|}\Big)$$

such that $v(\zeta) \le \int_{\mathbb{T}^n} vk \, \mathrm{d}m_n$ for every function v that is continuous on $\overline{\mathbb{D}}^n$ and plurisubharmonic on $\mathbb{D}^n$.

Proof. Let $\Theta(x) = \min_{\theta \in \mathbb{R}} J(e^{i\theta}x)$ for $x \in \mathbb{T}^n$. This is a continuous positive function. Let $c = \int \Theta \, \mathrm{d}m_n$ and $g = \Theta/c$. Then $\int g \, \mathrm{d}m_n = 1$, $g(e^{i\theta}x) = g(x)$ for every real θ, and $g \le J/c$. Define $k(y) = g(x)/J(x)$ for $y = \mu_\zeta(x)$. Then $0 \le k \le 1/c$, and by (3),

$$\int_{\mathbb{T}^n} h\, k \, \mathrm{d}m_n = \int_{\mathbb{T}^n} h(\mu_\zeta(x)) g(x) \, \mathrm{d}m_n(x) \tag{5}$$

for every bounded Borel function h; in particular $\int k \, \mathrm{d}m_n = 1$.

The Jensen inequality. Fix $x \in \mathbb{T}^n$ and let $\varphi_x(\lambda) = \mu_\zeta(\lambda x)$. This map is holomorphic in a neighbourhood of $\overline{\mathbb{D}}$, with $\varphi_x(0) = \zeta$ and $\varphi_x(\overline{\mathbb{D}}) \subset \overline{\mathbb{D}}^n$. Hence $v \circ \varphi_x$ is continuous on $\overline{\mathbb{D}}$ and subharmonic in $\mathbb{D}$, and

$$v(\zeta) \le \int_0^{2\pi} v(\mu_\zeta(e^{i\theta}x)) \frac{\mathrm{d}\theta}{2\pi}.$$

We integrate against $g(x) \, \mathrm{d}m_n(x)$ and use Fubini's theorem. The substitution $x' = e^{i\theta}x$ preserves m_n, and $g(e^{-i\theta}x') = g(x')$. By (5) we obtain

$$v(\zeta) \le \int_0^{2\pi} \int_{\mathbb{T}^n} v(\mu_\zeta(x')) g(x') \, \mathrm{d}m_n(x') \frac{\mathrm{d}\theta}{2\pi} = \int_{\mathbb{T}^n} v\, k \, \mathrm{d}m_n.$$

The bound. For $w \in \mathbb{D}$ we have

$$\log|1 + w| = \operatorname{Re} w + \operatorname{Re} \sum_{j \ge 2} \frac{(-1)^{j+1} w^j}{j} \le \operatorname{Re} w - \log(1 - |w|) - |w|.$$

Hence, for $a \in \mathbb{D}$ and $z \in \mathbb{T}$, with $s = |a|$,

$$\log J_a(z) \geq -2\operatorname{Re}(\bar{a}z) - \psi(s), \qquad \psi(s) = -\log(1+s) - 3\log(1-s) - 2s.$$

We have $\psi(0) = 0$ and

$$\psi'(s) = \frac{4s + 2s^2}{1 - s^2} \leq \frac{4s - 2s^2}{(1-s)^2} = \frac{\mathrm{d}}{\mathrm{d}s}\frac{2s^2}{1-s},$$

since after multiplication by $(1-s)^2(1+s)$ the inequality reads $4s - 2s^2 - 2s^3 \leq 4s + 2s^2 - 2s^3$. Hence $\psi(s) \leq 2s^2/(1-s)$. We apply this with $a = \zeta_j$ and $z = e^{i\theta}x_j$ and sum over j. Writing $M(x) = \sum_j \bar{\zeta}_j x_j$ and $\Sigma = \sum_j 2|\zeta_j|^2/(1-|\zeta_j|)$, we find that

$$\log J(e^{i\theta}x) \geq -2\operatorname{Re}(e^{i\theta}M(x)) - \Sigma \geq -2|M(x)| - \Sigma$$

for every real θ. The coordinate functions are orthonormal in $L^2(m_n)$, so $\int |M|^2 \,\mathrm{d}m_n = \tau^2$. By the convexity of the exponential function and the Cauchy–Schwarz inequality,

$$c \geq e^{-\Sigma}\int_{\mathbb{T}^n} e^{-2|M|}\,\mathrm{d}m_n \geq e^{-\Sigma}\exp\Bigl(-2\Bigl(\int_{\mathbb{T}^n}|M|^2\,\mathrm{d}m_n\Bigr)^{1/2}\Bigr) = e^{-2\tau-\Sigma}.$$

Since $k \leq 1/c$, this completes the proof. □

Lemma 1 is uniform in n, and a weak-star limit carries it to $\mathbb{T}^\infty$. This gives Theorem 2, whose necessity part is due to Cole and Gamelin.

Proof of Theorem 2. Suppose first that $k\,\mathrm{d}m$ is a representing measure for ζ with $k \in L^\infty(m)$. Then $k \in L^2(m)$, and for every $q \in \mathcal{P}_\infty$ the Cauchy–Schwarz inequality gives

$$|q(\zeta)| = \Bigl|\int_{\mathbb{T}^\infty} q\,k\,\mathrm{d}m\Bigr| \leq \|k\|_{L^2(m)}\|q\|_{L^2(m)}.$$

Hence the evaluation at ζ extends continuously to $H^2(\mathbb{T}^\infty)$, and $\zeta \in \ell^2$ by [3, Theorem 8.1], applied with $\mathrm{d}\omega = \mathrm{d}\theta/(2\pi)$ on $\mathbb{T}$ and $p = 2$. This is the necessity part of the L^p theorem of Cole and Gamelin [3, Section 1].

Conversely, suppose that $\zeta \in \mathbb{D}_2^\infty$. Let k_n be the density of Lemma 1 for $(\zeta_1, \dots, \zeta_n)$, regarded as a function on $\mathbb{T}^\infty$. The bound in Lemma 1 is at most $K(\zeta)$, so $0 \leq k_n \leq K(\zeta)$. Since $L^1(m)$ is separable, a subsequence converges weak-star in $L^\infty(m)$ to some k with $0 \leq k \leq K(\zeta)$ and $\int k\,\mathrm{d}m = 1$. Let $q \in \mathcal{P}_\infty$ depend on the first N variables, and let $L > 0$. The function $\max(\log|q|, -L)$ is continuous and plurisubharmonic on $\mathbb{C}^N$, and $q(\zeta^{(n)}) = q(\zeta)$ for $n \geq N$. By Lemma 1 and weak-star convergence, $\max(\log|q(\zeta)|, -L) \leq \int \max(\log|q|, -L)k\,\mathrm{d}m$. Letting $L \to \infty$, the monotone convergence theorem gives the Jensen inequality for q. If $f \in A$, choose $q_j \in \mathcal{P}_\infty$ with $|q_j| \leq |f| + \varepsilon_j$ on $\overline{\mathbb{D}}^\infty$ and $q_j(\zeta) \to f(\zeta)$, where ε_j decreases to 0. Then $\log|q_j(\zeta)| \leq \int \log(|f| + \varepsilon_j)k\,\mathrm{d}m$, and the monotone convergence theorem gives the Jensen inequality for f. Applying it to $e^{\pm f}$ and $e^{\pm if}$ shows that $k\,\mathrm{d}m$ represents ζ. □

Finally, suppose that $\zeta \in \overline{\mathbb{D}}^\infty$ and $|\zeta_j| = 1$ for some j, and let μ be a representing measure for ζ. The function $u = (1 + \bar{\zeta}_j z_j)/2$ belongs to A, satisfies $|u| \le 1$ on $\overline{\mathbb{D}}^\infty$ with equality only where $z_j = \zeta_j$, and $\int u \, \mathrm{d}\mu = u(\zeta) = 1$. Hence μ is carried by the set $\{z_j = \zeta_j\}$, which is m-null, so μ is singular to m.

4. Proof of Theorem 1

Lemma 1 gives part (1) at once. Part (2) then follows from the density of $\mathcal{P}_\infty$, and part (3) from a dilation argument.

Proof of Theorem 1 (1). Suppose that q depends on $z_1, \dots, z_n$. The function $v = \log(1+|q|)$ is continuous and plurisubharmonic on $\mathbb{C}^n$, since $t \mapsto \log(1+e^t)$ is convex and increasing and $\log|q|$ is plurisubharmonic. We apply Lemma 1 to $(\zeta_1, \dots, \zeta_n)$. The bound for k there is at most $K(\zeta)$, since the sums in (2) only grow when coordinates are added. As $v \ge 0$, we get $\log(1 + |q(\zeta)|) \le K(\zeta) \int v \, \mathrm{d}m_n = K(\zeta)\|q\|_0$. □

Proof of Theorem 1 (2). Fix $\zeta_0 \in \mathbb{D}_2^\infty$ and $0 < \delta < 1 - \|\zeta_0\|_\infty$, and let U be the ℓ^2-ball of radius δ centred at ζ_0. Then $U \subset \mathbb{D}_2^\infty$, and by the second bound in (2) we have $K \le K_0$ on U, where

$$K_0 = \exp\Big(2(\|\zeta_0\|_2 + \delta) + \frac{2(\|\zeta_0\|_2 + \delta)^2}{1 - \|\zeta_0\|_\infty - \delta}\Big).$$

Let $q_m \in \mathcal{P}_\infty$ with $\|q_m - f\|_0 \to 0$. By part (1), $\log(1 + |q_m(\zeta) - q_l(\zeta)|) \le K_0\|q_m - q_l\|_0$ for $\zeta \in U$, so (q_m) converges uniformly on U. Since $\|\cdot\|_0$ satisfies the triangle inequality, $\|q_m\|_0 \to \|f\|_0$, and part (1) gives $|q_m| \le \exp(K_0 \sup_l \|q_l\|_0)$ on U. Interlacing two approximating sequences shows that the limit F does not depend on the sequence. Hence F is locally bounded on $\mathbb{D}_2^\infty$, and on each complex line it is a locally uniform limit of holomorphic functions. So F is holomorphic on $\mathbb{D}_2^\infty$, and it satisfies the bound of part (2). On $\mathbb{D}_1^\infty$ it coincides with $\tilde{f}$, which is defined by the same limit (see the discussion before Proposition 2.25 in [4]). Holomorphic functions are continuous, and $\mathbb{D}_1^\infty$ is dense in $\mathbb{D}_2^\infty$, so the extension is unique. □

Proof of Theorem 1 (3). Let $0 < r < 1$. The map $\zeta \mapsto r \circ \zeta$ is linear and bounded from ℓ^2 to ℓ^1, and it maps $\overline{\mathbb{D}}^\infty$ into $\mathbb{D}_1^\infty$. Hence $G_r(\zeta) = F(r \circ \zeta)$ is holomorphic on $\mathbb{D}_2^\infty$. Moreover, $u_r(w) = \log(1 + |F(r \circ w)|)$ is continuous on $\overline{\mathbb{D}}^\infty$ with the product topology, since $w \mapsto r \circ w$ is continuous from $\overline{\mathbb{D}}^\infty$ to ℓ^1, and u_r is subharmonic in each variable.

A uniform bound. Fix $\zeta \in \mathbb{D}_2^\infty$ and $n \ge 1$. For w in the polydisc $\{w \in \mathbb{C}^n : |w_j| < 1/r\}$, which is a neighbourhood of $\overline{\mathbb{D}}^n$, the point $r \circ (w, 0) = (rw_1, \dots, r^n w_n, 0, \dots)$ lies in $\mathbb{D}_1^\infty$. The function $w \mapsto F(r \circ (w, 0))$ is continuous and separately holomorphic there, hence holomorphic in the n variables. Thus $v(w) = u_r(w, 0) = \log(1 + |F(r \circ (w, 0))|)$ is continuous and

plurisubharmonic in a neighbourhood of $\overline{\mathbb{D}}^n$, and Lemma 1 gives

$$u_r(\zeta^{(n)}) \le K(\zeta) \int_{\mathbb{T}^n} v \, \mathrm{d}m_n.$$

For fixed $(w_1, \dots, w_n) \in \mathbb{T}^n$, we apply (4) to u_r as a function of the remaining variables w', and find that $v(w) \le \int u_r(w, w') \, \mathrm{d}m(w')$. Hence $u_r(\zeta^{(n)}) \le K(\zeta) \int u_r \, \mathrm{d}m \le K(\zeta)\|F\|_0$. As $n \to \infty$, $r \circ \zeta^{(n)} \to r \circ \zeta$ in ℓ^1, and so

$$\log(1 + |G_r(\zeta)|) \le K(\zeta)\|F\|_0, \qquad \zeta \in \mathbb{D}_2^\infty, \ 0 < r < 1. \tag{6}$$

Passage to the limit. Let ζ_0, δ, U and K_0 be as in the proof of part (2), and let U' be the ℓ^2-ball of radius $\delta/4$ centred at ζ_0. By (6), $|G_r| \le M := \exp(K_0\|F\|_0)$ on U for every r. Let $x, y \in U'$ with $y \ne x$, and put

$$h(\lambda) = G_r\Big(x + \lambda \frac{\delta(y - x)}{2\|y - x\|_2}\Big), \qquad \lambda \in \mathbb{D}.$$

Then h is holomorphic and bounded by M in $\mathbb{D}$, so the Cauchy estimates give $|h'(\lambda)| \le M/(1 - |\lambda|)$. With $s = 2\|y - x\|_2/\delta \le 1$ we have $G_r(y) = h(s)$. If $s \le 1/2$, then $|G_r(y) - G_r(x)| = |h(s) - h(0)| \le 2Ms$, and if $s > 1/2$, then $|G_r(y) - G_r(x)| \le 2M < 4Ms$. Hence

$$|G_r(y) - G_r(x)| \le \frac{8M}{\delta}\|y - x\|_2, \qquad x, y \in U', \ 0 < r < 1,$$

so the functions G_r are equicontinuous on U'. For $\zeta \in \mathbb{D}_1^\infty$ we have $r \circ \zeta \to \zeta$ in ℓ^1 as $r \to 1$, so $G_r(\zeta) \to F(\zeta)$. Since $\mathbb{D}_1^\infty \cap U'$ is dense in U', equicontinuity shows that G_r converges pointwise on U' as $r \to 1$. The limit is bounded by M. On each complex line it is a pointwise limit of uniformly bounded holomorphic functions, hence holomorphic by Vitali's theorem. The limits on the balls U' therefore define a holomorphic function on $\mathbb{D}_2^\infty$. It extends F, it satisfies the bound of part (3) by (6), and it is unique by density. □

5. The domain $\mathbb{D}_2^\infty$ is maximal

We next show that Theorem 1 cannot hold beyond ℓ^2, already for the polynomials and the Hilbert space $H^2(\mathbb{T}^\infty)$.

Proposition 1. *Suppose that $\zeta \in \mathbb{D}^\infty \setminus \ell^2$. Then:*

1. *there are $q_N \in \mathcal{P}_\infty$ with $\|q_N\|_0 \to 0$ and $|q_N(\zeta)| \to \infty$;*
2. *there is $f \in H^2(\mathbb{T}^\infty) \subset N_*(\mathbb{T}^\infty)$ such that $\tilde{f}(\zeta^{(n)}) \to \infty$ as $n \to \infty$.*

Proof. Let $S_N = \sum_{j \le N} |\zeta_j|^2$, so that $S_N \to \infty$. Recall that $\log(1 + t) \le t$ for $t \ge 0$, so that $\|q\|_0 \le \|q\|_{L^1(m)} \le \|q\|_{L^2(m)}$.

(1) Let $q_N(w) = S_N^{-3/4} \sum_{j \le N} \bar{\zeta}_j w_j$. Then $\|q_N\|_0 \le \|q_N\|_{L^2(m)} = S_N^{-1/4} \to 0$, while $q_N(\zeta) = S_N^{1/4} \to \infty$.

(2) Let j_0 be the first index with $S_{j_0} > 0$, and let $a_j = \bar{\zeta}_j / S_j$ for $j \ge j_0$ and $a_j = 0$ otherwise. Since $S_j \to \infty$, the Abel–Dini theorem gives $\sum_j |\zeta_j|^2 / S_j^2 < \infty$ and $\sum_j |\zeta_j|^2 / S_j = \infty$. Hence $a \in \ell^2$, the function $f(w) = \sum_j a_j w_j$ belongs

to $H^2(\mathbb{T}^\infty)$, and $\tilde f(\zeta^{(n)}) = \sum_{j\le n}|\zeta_j|^2/S_j$ diverges. Here we used that $\tilde f$ is the pointwise limit of the partial sums on $\mathbb{D}_1^\infty$, by (1). □

Part (1) shows that no point outside ℓ^2 has a point evaluation that is continuous on $N_*(\mathbb{T}^\infty)$. Part (2) shows that if $\zeta \in \ell^p \setminus \ell^2$ for some $2 < p < \infty$, then $\tilde f$ has no extension to $\mathbb{D}_2^\infty \cup \{\zeta\}$ that is continuous at ζ for the ℓ^p norm, since $\zeta^{(n)} \to \zeta$ in ℓ^p; the same holds for the norm of c_0 when $\zeta \in c_0$. Both statements already hold for H^2, as in the work of Cole and Gamelin [3, Theorem 8.1].

6. Point evaluation for Dirichlet series

Let $\mathscr{N}_u$ be the class of Dirichlet series $f(s) = \sum_n a_n n^{-s}$ with abscissa of uniform convergence $\sigma_u(f) \le 0$ and

$$\|f\|_0 = \sup_{\sigma>0} \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \log(1+|f(\sigma+it)|)\,\mathrm{d}t < \infty.$$

This class was introduced by Brevig and Perfekt [2, Section 4]. The normalisation of $\|\cdot\|_0$ used here is that of Guo, Ni and Zhou [4, Section 4.1], who note that it is equivalent to the one in [2]; see also [1, Definition 1.1]. Let $\mathscr{N}$ be its completion in the metric $\|f-g\|_0$, and let $\mathscr{N}_* \subset \mathscr{N}$ be the closure of the Dirichlet polynomials [4, Section 4.1]. Let p_j be the j-th prime, and let $\alpha(n)$ be the multi-index with $n = \prod_j p_j^{\alpha_j}$. By [4, Theorem 4.1], for $f = \sum_n a_n n^{-s} \in \mathscr{N}_u$ the Bohr lift $\mathcal{B}f(\zeta) = \sum_n a_n \zeta^{\alpha(n)}$ converges on $\mathbb{D}_1^\infty$, belongs to $N(\mathbb{D}_1^\infty)$, and satisfies $\|\mathcal{B}f\|_0 = \|f\|_0$; hence $\mathcal{B}$ extends to an isometry from $\mathscr{N}$ into $N(\mathbb{D}_1^\infty)$.

For $s \in \mathbb{C}$ with $\sigma = \operatorname{Re} s > 1/2$ we put $r(s) = (p_j^{-s})_{j\ge1}$. Then $\|r(s)\|_2^2 = \sum_p p^{-2\sigma} < \infty$ and $\|r(s)\|_\infty = 2^{-\sigma} < 1$, so $r(s) \in \mathbb{D}_2^\infty$. Moreover, $r(s) \in \mathbb{D}_1^\infty$ if and only if $\sigma > 1$.

Corollary 1. *For $f \in \mathscr{N}$ let F be the extension of $\mathcal{B}f$ to $\mathbb{D}_2^\infty$ given by Theorem 1 (3), and put $f(s) = F(r(s))$ for $s \in \mathbb{C}_{1/2}$. Then f is holomorphic in $\mathbb{C}_{1/2}$. Suppose that $f, g \in \mathscr{N}$ and $\sigma = \operatorname{Re} s > 1/2$. Then*

$$\log(1+|f(s)-g(s)|) \le K(r(s))\|f-g\|_0,$$

where

$$\begin{aligned} K(r(s)) &= \exp\Big(2\Big(\sum_p p^{-2\sigma}\Big)^{1/2} + \sum_p \frac{2p^{-2\sigma}}{1-p^{-\sigma}}\Big) \\ &\le \exp\Big(2\sqrt{\log\zeta(2\sigma)} + \frac{2\log\zeta(2\sigma)}{1-2^{-\sigma}}\Big). \end{aligned}$$

In particular, $f \mapsto f(s)$ is continuous on $\mathscr{N}$ and on $\mathscr{N}_*$ for every $s \in \mathbb{C}_{1/2}$. If $f = \sum_n a_n n^{-s} \in \mathscr{N}_u$, then $f(s)$ is the sum of the series for every $s \in \mathbb{C}_{1/2}$.

Proof. The map $r\colon \mathbb{C}_{1/2} \to \ell^2$ is holomorphic, being the locally uniform limit of its truncations, and F is holomorphic on $\mathbb{D}_2^\infty$. Hence $f = F \circ r$ is holomorphic in $\mathbb{C}_{1/2}$. The extensions of $\mathcal{B}f$, $\mathcal{B}g$ and $\mathcal{B}(f-g)$ satisfy the linear

relation on $\mathbb{D}_1^\infty$, hence on $\mathbb{D}_2^\infty$ by uniqueness, and $\|\mathcal{B}(f-g)\|_0 = \|f-g\|_0$. So the inequality is Theorem 1 (3) at $\zeta = r(s)$. The formula for $K(r(s))$ follows from (2) with $|\zeta_j| = p_j^{-\sigma}$. The bound follows from $1-p^{-\sigma} \geq 1-2^{-\sigma}$ and

$$\sum_p p^{-x} \leq \sum_p \sum_{k\geq 1} \frac{p^{-kx}}{k} = \log\zeta(x), \qquad x > 1.$$

Finally, let $f \in \mathscr{N}_u$. Since $\sigma_u(f) \leq 0$, Bohr's theorem shows that the series converges absolutely in $\mathbb{C}_{1/2}$, so its sum is holomorphic there. For $s \in \mathbb{C}_1$ the point $r(s)$ lies in $\mathbb{D}_1^\infty$, where $F(r(s)) = \mathcal{B}f(r(s)) = \sum_n a_n n^{-s}$. By the identity theorem, the two holomorphic functions agree in $\mathbb{C}_{1/2}$. □

The half-plane $\mathbb{C}_{1/2}$ is optimal: for no s with $\sigma = \operatorname{Re} s \leq 1/2$ is the evaluation at s continuous on the Dirichlet polynomials for $\|\cdot\|_0$. Indeed, let $A_N = \sum_{n\leq N} n^{-2\sigma}$, so that $A_N \to \infty$, and let $q_N(w) = A_N^{-3/4} \sum_{n\leq N} n^{-\bar{s}} n^{-w}$. By [4, (4.1)] and $\log(1+t) \leq t$, we have $\|q_N\|_0 \leq \|\mathcal{B}q_N\|_{L^2(m)} = A_N^{-1/4} \to 0$, while $q_N(s) = A_N^{1/4} \to \infty$.

7. Concluding remarks

Other Jensen measures. The invariance $g(e^{i\theta}x) = g(x)$ in the proof of Lemma 1 is used only to remove the average over θ in the Jensen step. For an arbitrary probability density g on $\mathbb{T}^n$, the same computation shows that $h \mapsto \int h(\mu_\zeta(x)) g_{\mathbb{T}}(x)\,\mathrm{d}m_n(x)$ is a Jensen measure for ζ, where

$$g_{\mathbb{T}}(x) = \int_0^{2\pi} g(e^{i\theta}x) \frac{\mathrm{d}\theta}{2\pi}$$

is the average of g over the circle through x. The choice $g = 1$ gives the Poisson measure, whose density $1/J(\mu_\zeta^{-1}(y))$ has supremum $\prod_j (1+|\zeta_j|)/(1-|\zeta_j|)$. Among circle-invariant g, the choice $g = \Theta/c$ in Lemma 1 minimises the supremum of the density of the pushed-forward measure. Indeed, its density at $\mu_\zeta(e^{i\theta}x)$ is $g(x)/J(e^{i\theta}x)$, which equals $g(x)/\Theta(x)$ for a suitable θ, and $\sup g/\Theta \geq \int g\,\mathrm{d}m_n / \int \Theta\,\mathrm{d}m_n = 1/c$.

Other product measures. Cole and Gamelin [3, Section 8] also consider the product measures $\mathrm{d}Q = \mathrm{d}\omega \times \mathrm{d}\omega \times \cdots$ on $\overline{\mathbb{D}}^\infty$, where ω is a rotation-invariant probability measure on $\overline{\mathbb{D}}$ other than the point mass at 0, and ρ_0 is the norm of z in $L^\infty(\omega)$. Our construction uses the change of variables (3), which is special to the Haar measure of $\mathbb{T}$, and we do not address the following question [3, p. 130].

Problem 1 (Cole and Gamelin)**.** *Suppose that $\zeta \in \ell^2$ and $|\zeta_j| < \rho_0$ for every j. Is there a positive representing measure for ζ which is absolutely continuous with respect to Q, with bounded density?*

The size of the constant. When $\|\zeta\|_\infty \le 1/2$, we have $\log K(\zeta) \le 2\|\zeta\|_2 + 4\|\zeta\|_2^2$, while $\log\|P_\zeta\|_\infty \ge 2\|\zeta\|_1$. The gain over (1) is therefore large when $\|\zeta\|_2$ is much smaller than $\|\zeta\|_1$; as noted in the introduction, it is a loss at points with a single nonzero coordinate.

Let $C(\zeta)$ denote the smallest constant such that $\log(1+|F(\zeta)|) \le C(\zeta)\|F\|_0$ for every $F \in N(\mathbb{D}_1^\infty)$. Testing with $F(w) = t\sum_j \bar\zeta_j w_j$, for which $\|F\|_0 \le t\|\zeta\|_2$ and $F(\zeta) = t\|\zeta\|_2^2$, and letting $t \to 0$, we find that $C(\zeta) \ge \|\zeta\|_2$. Next, fix j with $\zeta_j \ne 0$, put $s = |\zeta_j|$ and $u = \zeta_j/s$, and let $F(w) = G(w_j)$, where

$$G(z) = \exp\Big(\lambda\frac{u+z}{u-z}\Big), \qquad \lambda > 0.$$

Then $\log|G(z)| = \lambda(1-|z|^2)/|u-z|^2$ is λ times the Poisson kernel at u. It is nonnegative, and its mean over every circle $|z| = \rho < 1$ equals λ. Since $\log(1+e^x) \le \log 2 + x$ for $x \ge 0$, we get $\|F\|_0 \le \lambda + \log 2$, while $\log(1+|F(\zeta)|) \ge \log|G(\zeta_j)| = \lambda(1+s)/(1-s)$. Letting $\lambda \to \infty$, we find that $C(\zeta) \ge (1+s)/(1-s)$. Since $\zeta \in c_0$, the largest $|\zeta_j|$ is attained, and hence

$$C(\zeta) \ge \max\Big\{\|\zeta\|_2, \max_j \frac{1+|\zeta_j|}{1-|\zeta_j|}\Big\} = \max\Big\{\|\zeta\|_2, \frac{1+\|\zeta\|_\infty}{1-\|\zeta\|_\infty}\Big\}.$$

By the computation in the introduction, $K(\zeta)$ is indeed larger than $(1+\|\zeta\|_\infty)/(1-\|\zeta\|_\infty)$.

For $t > 0$ let

$$C^*(t) = \sup\big\{C(\zeta);\ \zeta \in \mathbb{D}_2^\infty,\ \|\zeta\|_2 \le t,\ \|\zeta\|_\infty \le 1/2\big\}.$$

By the bounds above, $\max\{t, 1\} \le C^*(t) \le \exp(2t + 4t^2)$.

Problem 2. *Determine the order of growth of $C^*(t)$ as $t \to \infty$.*

In particular, we do not know whether the exponential dependence on $\|\zeta\|_2$ in (2) is necessary.

Problem 3. *Is C^* polynomially bounded, that is, is there a polynomial Q such that $C^*(t) \le Q(t)$ for every $t > 0$?*

A positive answer to Problem 3, or to its analogue with $1/2$ replaced by $2^{-1/2}$, would improve the bound of Corollary 1 near the line $\operatorname{Re} s = 1/2$, where $\log\zeta(2\sigma)$ is of order $\log(1/(2\sigma-1))$.

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